

Results

01 • CHI-SQUARE GOODNESS-OF-FIT

do counts match an expected split?

Table 1. Observed vs. expected

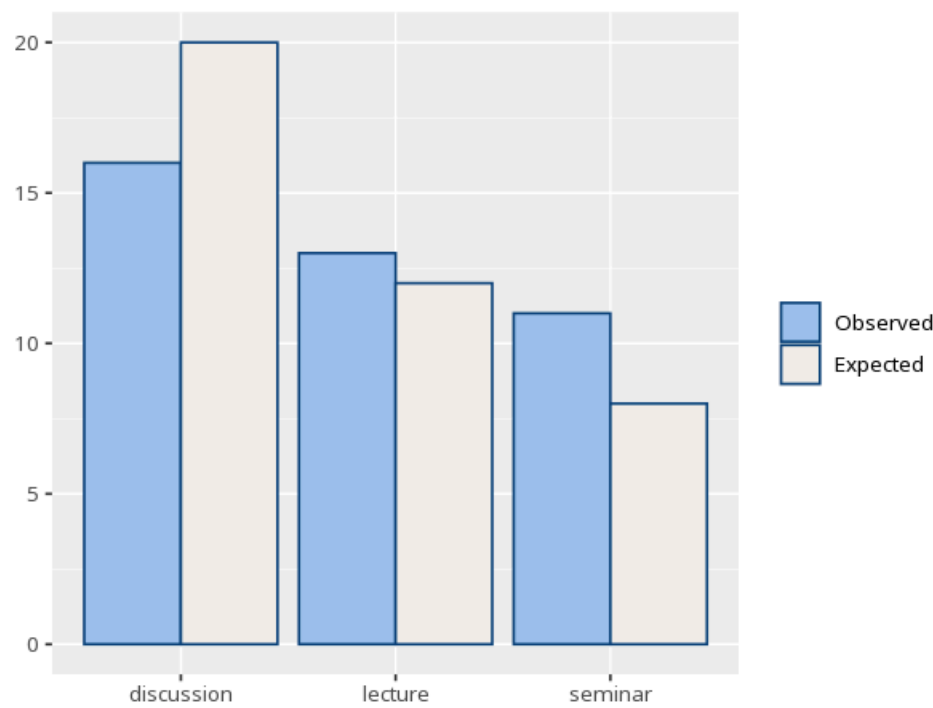
Category	Observed	Expected	Std. residual
discussion	16	20.00	-1.26
lecture	13	12.00	0.35
seminar	11	8.00	1.19

Table 2. Goodness-of-fit test

χ^2	df (k-1)	p	Cohen's w [95% CI]
2.01	2	.366	0.22 [0.00, 2.00]

here df = number of categories – 1 (not (r-1)(c-1) as in the independence test).

Figure. Observed vs. expected



How to read this test

Tests whether one categorical variable's counts depart from an expected distribution (equal by default). A p below alpha means the observed split differs from expected; a standardized residual beyond about ± 1.96 flags a category that significantly drives the result. Valid only when expected counts are mostly ≥ 5

(some texts allow ≥ 1 if no more than 20% are below 5); for sparse categories use an exact / simulated test instead. Your significance threshold (α) is 0.05.

APA template: A goodness-of-fit test, $\chi^2(2, N=40)=2.01$, $p = .366$, $w=.22$ [.00, 2.00].

After export · optional feedback — an anonymous, optional satisfaction survey — opens a short Google Form in a new tab

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